

2026-06-14 The first short proposal

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Outline

- ① The story
- ② Some issues to be discussed
- ③ Related literature

What is the best part of LinkedIn data

With Revelio lab data, we know the lifecycle of a vacancy.

- From job posting data, we have information on the creation of a vacancy.
 - Full job advertisements in text.
 - Some standardized variables: Company, job title, location, salary range (if existing).
- From individual profiles data, we have information on who filled in the vacancy.
 - Education, gender, race (estimated), location, etc.
 - Past and future work experiences: In particular, subsequent career progression (promotion, exit, occupation transition, etc).

The story in mind: Different hiring strategies

Consider a sector where asymmetric information about workers' productivity matters a lot¹. There could be two potential strategies for talent discovery:

- Screening-first strategy:
 - High job requirements, more tests and interviews.
 - The aim is to improve the average productivity (and reduce the variance) of the new hires.
- Hiring-first strategy:
 - The firm may spend fewer resources to screen workers before hiring (due to, e.g., high screening costs, low screening precision).
 - Instead, through employer learning on the job, the firm may fire unqualified employees with low productivity and keep only those with high productivity.

¹For example, medicine companies hiring entry-level researchers to develop new drugs.

Part 1 of the paper: Classification

Classification $\implies$ Validation $\implies$ Implications

- The first part is to classify firms by their talent discovery strategies.
- The rough idea is as follows.
 - We construct a series of variables from job posting and individual profiles (some textual analysis is useful).
 - Then we classify firms into different clusters with some data-driven methods (essentially an unsupervised machine learning problem).
 - Even though we don't know firms' real hiring practices, hopefully, we can open this black box through analysis on the creation and filling of vacancies.

Part 2 of the paper: Validation

Classification $\implies$ Validation $\implies$ Implications

- The hiring strategy of a firm is not random – it is already optimized based on its market conditions.
- In particular, the productivity distribution of potential applicants and firing costs will be important factors to shape a firm's hiring strategy.
- We may expect firms differ systematically in their hiring strategies in certain ways.

Part 3 of the paper: Implications

Classification $\implies$ Validation $\implies$ Implications

- What are the implications if firms have different hiring strategies?
- We could collect external objective productivity measures (e.g., patents, clinical trials results) for firms or ideally, for individuals.
- With this productivity measure, we could investigate how firms' hiring strategies affect (1) sorting patterns between employees and firms; (2) firm productivity; (3) economy-wide productivity and efficiency.

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The occupation(s) of interest

- The first big decision is to decide which occupations we are investigating.
- Two criteria:
 - Asymmetric information matters: Entry-level jobs instead of senior hires (where more signals on productivity are available).
 - Available objective productivity measures: Patents, lawsuits results, clinical trials, etc.
- The best approach may be to (1) first pick some good candidates occupations², then (2) investigate the coverage patterns in LinkedIn data and the presence of objective productivity measures.

²Some anecdotal evidence about how different firms in the same sector hire workers will be super helpful.

A data-driven method to classify firms by their talent discovery strategies

This will be the biggest contribution to the literature and yet the most difficult part.

- Contributions:
 - We only know what kind of people a vacancy wants (from job posting), who actually fills in the vacancy and their subsequent career trajectories (from individual profiles).
 - Even though we don't know how firms actually make the hiring decisions, we hope to say something about it in a data-driven way.
- Difficulties:
 - Suppose we have constructed a dataset about a universe of firms hiring the same occupation over years 2010-18. Each firm has multiple vacancies over the years; in each vacancy, multiple employees join the firm; for each employee, we observe his career path.
 - Difficulty 1: *We need to decide which variables to construct (and how to construct them from text) with these two sources.*
 - Difficulty 2: *We need to design an algorithm to aggregate these variables at different levels to a firm-level variable denoting their hiring strategies.*

Next steps

- For the LinkedIn data, some descriptive results about (1) coverage patterns for the occupation(s) of interest; (2) the matching from individual profiles to job posting (so that we know who fills in the vacancy); (3) job advertisements text.
- For the external productivity data, some background information about (1) the sources and context; (2) how to link them to individuals or firms.

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Hiring practices

- Two chapters in the *Handbook of Labor Economics*: Hoffman and Stanton, 2025; Paul and Scott, 2011.
- Specific hiring practices/technologies:
 - Hiring as a bandit problem (exploration vs exploitation): Lazear, 1998; Li et al., 2026.
 - Referrals: Friebe et al., 2023; Hensvik and Skans, 2016; Pallais and Sands, 2016.
 - Screening tests: Autor and Scarborough, 2008; Hoffman, Kahn, et al., 2018; Mocanu, 2024.
 - Signals/Information: Agan et al., 2025; Bryan et al., 2026; Stanton and Thomas, 2016
- Not so much research (if any) on different hiring practices and their consequences across firms.

Data-driven method to uncover unobserved heterogeneity and practices

The following 4 papers are far from complete. We need to do extensive literature review on this part.

- Classifying individual heterogeneity: Castro et al., 2026; Gregory et al., 2025
- Describing internal labor markets: Ewens and Giroud, 2025; Huitfeldt et al., 2023

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